

SomaTrack: Implementation Pipeline Specification

Predicting Study-Related Physical Ailments using ML

Executive Summary

Project Goal: Build a multiclass classification model that predicts physical discomfort levels (None, Mild, Moderate, Chronic) based on study habits and environmental factors.

Current Status: 270 responses collected across 34 survey questions

Primary Targets: Back Pain, Neck Strain, Tension Headaches, Eye Strain, Wrist Pain

Model Type: Multiclass Classification

Success Metric: F1-Score ≥ 0.75 for primary pain categories

Phase 1: Data Preparation & Cleaning

1.1 Initial Data Assessment

- **Current Dataset:** 270 responses, 34 features
- **Missing Values:** Minimal (Q10: 2, Q18: 1, Demographics: 5-10)
- **Action:** Document missing data patterns; decide on imputation strategy

1.2 Feature Engineering & Encoding

Numerical Features (Already Present)

- Q1: Hours spent sitting per day
- Q3: Longest continuous sitting time (hours)
- Q9: Daily screen time (hours)
- Q29: Age
- Q2: Study days per week

Actions:

- Normalize/standardize these features (0-1 scale or z-score)
- Check for outliers using IQR or z-score methods
- Handle any missing values with median imputation

Categorical Features to Encode

Feature	Question	Encoding Strategy
Study Location	Q11	One-Hot or Label Encoding (Library, Bed, Desk, Café)
Seat Type	Q12	One-Hot (Plastic chair, Stool, Ergonomic chair)
Input Method	Q13	Binary/One-Hot (Trackpad vs External peripherals)
Posture	Q14	One-Hot (Good, Neutral, Poor)
Breaks Taken	Q4	Binary (Yes/No)
Leave Desk During Break	Q6	Binary (Yes/No)
Screen at Eye Level	Q16	Binary (Yes/No)
Lean on Back	Q15	Binary (Yes/No)
Medical History	Q17	Binary (Yes/No)
Exercise Weekly	Q20	Binary (Yes/No)
Gender	Q30	Binary/One-Hot
Institution Type	Q31	One-Hot (if multiple)
Field of Study	Q32	One-Hot (if too many, group rare categories)
Year of Study	Q33	Ordinal encoding (1st, 2nd, 3rd, 4th+)
Lighting	Q19	Ordinal (Poor, Moderate, Good)
Stress Level	Q10	Ordinal (Low, Medium, High)

Ordinal Features (Extract numeric scales):

- Q5: Break duration → convert to minutes/category
- Q7: Water consumption → extract numeric value
- Q8: Caffeine consumption → ordinal scale
- Q18: Backpack weight → extract numeric or ordinal
- Q21: Sleep duration → extract hours (numeric)

1.3 Target Variable Creation

Option A: Multiclass from Individual Pain Questions (Q22-Q27)

- Create separate models for each pain type:
 - Back Pain (Q22)
 - Neck Strain (Q23)
 - Tension Headaches (Q24)
 - Wrist Pain (Q25)
 - Eye Strain (Q26)
 - Finger Numbness (Q27)
- Each encoded as: 0=Never, 1=Rarely, 2=Sometimes, 3=Often, 4=Always

Option B: Composite Overall Discomfort (Q28)

- Use Q28 as primary target (already captures overall discomfort level)
- Create classes: None, Mild, Moderate, Chronic

Recommendation: Start with **Option A** (individual models) as it provides interpretability and allows identifying pain-specific risk factors. Later aggregate for overall discomfort model.

1.4 Data Quality Checks

Steps:

1. Check class distribution for imbalance (stratification concern)
2. Remove or flag suspicious responses (e.g., 0 hours study but high pain)
3. Verify consistency (e.g., if no breaks taken, break duration should be N/A)
4. Handle outliers in sitting hours, screen time, age
5. Create data quality report

Phase 2: Exploratory Data Analysis (EDA)

2.1 Univariate Analysis

- **Distribution plots:** Sitting hours, screen time, stress levels, pain frequencies
- **Histograms/KDE:** Numeric features
- **Bar charts:** Categorical features (location, seat type, exercise habits)

2.2 Bivariate Analysis - Pain Distribution by Key Factors

Visualizations to create:

- Back Pain by Study Location (Box plot + Count)
- Neck Pain by Seat Type (Bar chart with percentages)
- Headaches by Screen Time (Scatter plot with trendline)
- Overall Pain Distribution (Pie + Bar charts)
- Pain by Exercise Habits (Bar chart)
- Pain by Posture Quality (Heatmap)
- Pain by Stress Level (Box plot)

2.3 Correlation Analysis

- **Heatmap:** Feature correlations with pain outcomes
- **Point-biserial correlation:** Numeric vs pain levels
- **Cramér's V:** Categorical vs categorical associations
- Identify multicollinearity (e.g., sitting hours vs screen time)

2.4 Missing Data Visualization

- Pattern analysis of missing values
- Decision: Impute or drop responses with missing targets

2.5 Class Balance Analysis

- **Action:** Check if pain classes are balanced
- **If imbalanced:** Document ratio (may need SMOTE, class weights, or stratified k-fold)

Deliverable: Jupyter notebook with 15-20 publication-ready visualizations

Phase 3: Feature Selection & Engineering

3.1 Feature Importance Assessment

Methods:

1. **Correlation with targets** (point-biserial, Cramér's V)
2. **Mutual Information:** Categorical-target relationships
3. **Permutation Importance:** Post-model feature importance
4. **RFE (Recursive Feature Elimination):** Iterative removal

3.2 Feature Engineering

Create new features if beneficial:

- Sitting-to-break ratio: (continuous sitting) / (break frequency)
- Activity score: (exercise + leave desk) vs (sitting hours)
- Ergonomic risk index: (poor posture) + (non-ergonomic chair) + (screen not at eye level)
- Hydration index: water consumption vs sitting hours
- Stress-pain correlation: stress level × sitting hours

3.3 Dimensionality Reduction (if needed)

- PCA for numeric features if correlation is very high
- Or simple selection: keep top 15-20 most important features

Deliverable: Feature ranking report with justification for each included feature

Phase 4: Model Development & Training

4.1 Baseline Models (Compare Performance)

- **Logistic Regression** (Multinomial for multiclass)
- **Random Forest Classifier** (non-linear patterns)
- **Gradient Boosting** (XGBoost, LightGBM)
- **SVM with RBF kernel** (non-linear classification)
- **Neural Network** (MLP: optional, if dataset supports)

4.2 Train-Test Split Strategy

- Stratified Split: 80% train, 20% test
- Maintain class distribution in both sets
- Document random seed for reproducibility

4.3 Hyperparameter Tuning

- For each model:
 - GridSearchCV or RandomizedSearchCV
 - Cross-validation: Stratified 5-Fold
 - Optimize for: F1-Score (macro or weighted, depending on class imbalance)
 - Log all tuning experiments

4.4 Regularization & Class Handling

- **Class weights:** Adjust for imbalanced data
- **Stratified K-Fold:** Ensure valid cross-validation
- **Threshold tuning:** For binary conversion (if needed)

Deliverable: Model comparison table with metrics

Phase 5: Model Evaluation & Validation

5.1 Evaluation Metrics (per pain type)

For each classification model:

- Precision: True Positives / (TP + FP)
- Recall: True Positives / (TP + FN)
- F1-Score: Harmonic mean of Precision & Recall
- Confusion Matrix: Visualize classification errors
- ROC-AUC (if binary version): Probability ranking
- Classification Report: Macro/Weighted averages

5.2 Stratified K-Fold Cross-Validation

Steps:

1. Perform 5-Fold stratified cross-validation
2. Record metrics for each fold
3. Report mean $\pm$ std of each metric
4. Check for variance: high variance = potential overfitting/underfitting

5.3 Model Interpretation (White-box AI)

For best model(s):

- **SHAP values:** Feature importance + direction of impact
- **LIME:** Local interpretable explanations
- **Decision tree visualization** (if using tree-based model)
- **Coefficient analysis** (Logistic Regression)
- **PDP plots:** Partial dependence on key features

5.4 Error Analysis

- Analyze misclassifications: which pain levels are confused?
- Identify patterns in wrong predictions
- Check if certain subgroups (e.g., high-stress students) are systematically misclassified

Deliverable: Classification report with confusion matrices + interpretation report

Phase 6: Risk Factor Identification (Domain Insights)

6.1 Key Findings to Extract

Identify at least 3 major risk factors per pain type:

Example Output Format:

Back Pain - Top Risk Factors:

1. Sitting > 6 hours/day (F1-impact: 0.35)
2. Non-ergonomic chair (F1-impact: 0.28)
3. No exercise (F1-impact: 0.22)

Neck Strain - Top Risk Factors:

1. Screen not at eye level (F1-impact: 0.40)
2. Trackpad use without external mouse (F1-impact: 0.30)
3. Poor posture (F1-impact: 0.25)

Headaches - Top Risk Factors:

1. High stress + >8 hours sitting (interaction: 0.38)
2. Poor lighting (F1-impact: 0.32)
3. <6 hours sleep (F1-impact: 0.28)

6.2 Actionable Recommendations

For each risk factor, derive student-friendly advice:

- "Students with your profile reduced pain by 40% by moving to a desk"
 - "Using an external mouse could reduce neck strain risk by 30%"
 - etc.
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Phase 7: Deployment & Demo Application

7.1 Web Application / CLI Tool

Input Module:

- User enters their study habits
- Form fields for: sitting hours, location, seat type, posture, stress, etc.
- Simple Streamlit or Flask interface

Processing:

- Feature extraction & encoding
- Preprocessing (scaling, encoding)
- Load trained model

Output Module:

- Predicted pain risk level per category
- Risk score (0-100%)
- Top 3 modifiable risk factors for this user
- Personalized recommendations

7.2 Explainability Dashboard

- Feature importance chart
- Sample predictions with explanations
- Risk factor breakdown

Deliverable: Working demo (Streamlit app or Flask web interface)

Phase 8: Documentation & Presentation

8.1 Deliverables Checklist

- ☐ **SomaTrack Dataset:** Clean CSV (270+ responses, engineered features)
- ☐ **EDA Notebook:** Jupyter notebook with 20+ visualizations
- ☐ **Feature Engineering Report:** Feature selection justification
- ☐ **Model Training Notebook:** All experiments, hyperparameter tuning logs
- ☐ **Evaluation Report:** Metrics, confusion matrices, cross-validation results
- ☐ **Interpretation Report:** SHAP/LIME analysis, risk factors (white-box AI)
- ☐ **Trained Models:** Serialized model files (pkl/h5) for each pain type
- ☐ **Deployment Package:** Demo app (Streamlit/Flask/CLI)
- ☐ **Final Report:** Summary of findings, limitations, recommendations
- ☐ **Presentation Slides:** 15-20 slides with key results & live demo

8.2 Report Structure (Final)

1. Executive Summary (1 page)
 - Problem statement, approach, key findings
2. Literature Review / Problem Context (2 pages)
 - Why ergonomic health matters
 - Prevalence of study-related pain in universities
3. Methodology (3 pages)
 - Data collection, feature engineering, model selection
4. Results & Analysis (5 pages)
 - Best model performance
 - Top risk factors with visualizations
 - Interpretation of model decisions
5. Risk Factor Deep-Dive (2 pages)
 - 3+ major findings with actionable recommendations
6. Limitations & Future Work (1 page)
 - Model limitations, dataset bias, next steps
7. Appendix
 - All figures/tables
 - Code snippets (key sections)

8.3 Presentation Format

- **Live Demo:** 5 min (user enters data → model predicts pain risk)
 - **Results Walkthrough:** 10 min (visualizations, risk factors)
 - **Technical Depth:** 5 min (model architecture, validation approach)
 - **Insights & Recommendations:** 5 min (what can students do?)
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Technical Stack Recommendations

Programming & ML Libraries

```
Python 3.9+
pandas, numpy           # Data manipulation
scikit-learn            # ML models, preprocessing, metrics
xgboost, lightgbm       # Gradient boosting
matplotlib, seaborn     # Visualization
plotly                  # Interactive plots
jupyter                  # Notebooks
shap, lime               # Model interpretability
```

Deployment

```
Option 1: Streamlit (simplest)
- streamlit_app.py → one-file deployment

Option 2: Flask (more control)
- app.py, templates/, static/

Option 3: CLI Tool
- Python script with argparse
```

Version Control

- Git repository with clear folder structure
- Document data versioning (survey_responses_v1.csv, etc.)

Timeline & Milestones

Phase	Milestone	Duration
1	Data Cleaning & Encoding	1-2 weeks
2	EDA & Visualizations	1-2 weeks
3	Feature Engineering	1 week
4	Model Training & Tuning	2 weeks
5	Evaluation & Validation	1-2 weeks
6	Risk Factor Analysis	1 week
7	Demo Application	1-2 weeks
8	Documentation & Presentation	1-2 weeks
Total		10-14 weeks

Success Criteria (Checkpoints)

- ☐ **Phase 2:** 15+ quality visualizations showing pain distributions
- ☐ **Phase 4:** Best model achieves $F1 \geq 0.70$ on 5-fold CV
- ☐ **Phase 5:** Classification reports show balanced metrics (Precision $\approx$ Recall for main classes)
- ☐ **Phase 6:** Identify 3+ risk factors per pain type with statistical support
- ☐ **Phase 7:** Demo app successfully predicts and explains results
- ☐ **Phase 8:** Final presentation includes live demo + 5+ actionable insights

Potential Challenges & Mitigation

Challenge	Mitigation
Class Imbalance (e.g., few "Chronic Pain" cases)	Use SMOTE, class weights, or stratified k-fold; optimize F1 macro
Multicollinearity (e.g., sitting hours vs screen time)	Use VIF analysis; feature selection or PCA
Overfitting (model memorizes patterns)	Regularization, cross-validation, early stopping in boosting
Limited Dataset (270 responses)	Use ensemble methods, cross-validation; avoid deep learning
Feature Encoding Errors	Validate encoded features match survey responses; unit tests
Model Interpretability Trade-off	Prefer tree-based (Random Forest, XGBoost) over neural networks
Survey Response Bias	Document assumptions; recommend future sampling from diverse programs

Next Steps

- Start with Phase 1:** Load data, identify missing values, begin encoding
- Validate Features:** Ensure encoded features make sense (spot-check a few rows)
- Build EDA Notebook:** Create publication-ready visualizations
- Iteratively develop:** Models $\rightarrow$ evaluate $\rightarrow$ insights
- Document as you go:** Keep research log of decisions & findings

Appendix: Survey Question Mapping

Numeric Features

- Q1: Sitting hours/day
- Q2: Study days/week
- Q3: Max continuous sitting (hours)
- Q9: Daily screen time (hours)
- Q29: Age

Categorical/Ordinal Features

- Q4: Breaks? (Y/N)
- Q5: Break duration
- Q6: Leave desk? (Y/N)
- Q7: Water (liters)
- Q8: Caffeine frequency
- Q10: Stress level (Low/Med/High)
- Q11: Study location
- Q12: Seat type
- Q13: Input method
- Q14: Posture quality
- Q15: Lean on back? (Y/N)
- Q16: Screen at eye level? (Y/N)
- Q17: Medical history? (Y/N)
- Q18: Backpack weight
- Q19: Lighting quality
- Q20: Exercise weekly? (Y/N)
- Q21: Sleep hours/night
- Q30: Gender
- Q31: Institution type
- Q32: Field of study
- Q33: Year of study

Target Variables

- Q22: Back pain frequency
 - Q23: Neck strain frequency
 - Q24: Headache frequency
 - Q25: Wrist pain frequency
 - Q26: Eye strain frequency
 - Q27: Finger numbness frequency
 - Q28: Overall discomfort level
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